

Biology Background: What This Project Is Really Doing

A plain-English companion to the 15-day plan. No fancy words — where a real term is needed, it's explained right away. Reread this anytime you feel lost about *why* a step exists.

The big picture

Your cells contain DNA — the full instruction manual for the body. But DNA just sits in the nucleus; it doesn't do anything by itself. When a cell wants to *use* an instruction, it makes a working copy of that gene. That copy is called **RNA**. So the flow is:

DNA → RNA → protein (proteins are the machines that do the work in a cell).

The amount of RNA a cell makes for each gene tells you how "switched on" that gene is right now. If a diabetic person's cells make a lot more RNA for gene X than a healthy person's cells, that's a clue that gene X is involved in diabetes. **Measuring all the RNA in a sample is called RNA-seq.** That is the heart of this whole project.

The basic question the project asks:

"Which genes behave differently in diabetic islet cells compared to healthy ones — and can I map the hidden control system making them behave that way?"

The three types of RNA you care about

Not all RNA becomes protein. Three kinds matter here:

- **mRNA (messenger RNA)** — the "normal" kind. The working copy of a gene that gets turned into a protein. When people say "this gene is expressed," they usually mean its mRNA. These are the main **workers**.
 - **lncRNA (long non-coding RNA)** — long RNA that does *not* become a protein. Many of them are regulators: they help control which other genes are on or off. Think **managers**.
 - **miRNA (microRNA)** — very tiny RNA pieces whose job is to *shut genes down*. A miRNA finds a matching mRNA and silences it, so no protein gets made. Think **off-switches / brakes**.
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The ceRNA idea (the sponge story)

miRNAs (the off-switches) work by grabbing matching RNA. Normally they grab mRNAs and silence them. But lncRNAs can have the *same* matching spots that mRNAs do. So a lncRNA can act like a **sponge** — soaking up miRNAs and keeping them busy, so those miRNAs can't reach the mRNAs they were meant to silence.

Analogy: a predator (miRNA) eats a certain prey (mRNA). Drop in decoys (lncRNA) that look like prey. The predator wastes time chasing decoys, and the real prey survives — so more of that mRNA survives and makes more protein.

This chain — **lncRNA soaks up miRNA, which frees up mRNA** — is a **ceRNA network** (ceRNA = "competing endogenous RNA," just "RNAs competing for the same miRNAs"). Three layers:

lncRNA → (soaks up) → miRNA → (would have silenced) → mRNA

In diabetes these control chains can go wrong. Mapping them can reveal *why* certain genes misbehave. That map — the ceRNA network — is your final deliverable, and exactly what the job asks for.

Why this dataset (diabetic islets)

The samples come from **pancreatic islets** — clusters of cells in the pancreas that make **insulin**, the hormone that controls blood sugar. In type-2 diabetes these cells stop working properly. Comparing diabetic vs healthy islets is one of the most direct ways to see diabetes at the RNA level.

What each step gives you, and why

Raw data (FASTQ files). The sequencer doesn't say "gene X = 500 copies." It hands you millions of tiny RNA fragments, each written as letters (A, T, G, C), in files called **FASTQ**. On their own they're meaningless — like a shredded book on the floor. The next steps reassemble the book and count the pages.

Quality control + trimming (FastQC / fastp). Some fragments are low-quality or have leftover lab chemicals ("adapters") stuck on. This checks quality and snips off the bad bits. *Yield:* clean reads + a before/after report. (Cleaning smudges off the shreds.)

Alignment (HISAT2). Figure out *where in the human genome* each clean fragment came from — matching each shred back to its page. Output is a **BAM file** (every read + its location). *Yield:* you now know which gene each fragment belongs to.

Counting (featureCounts). Tally how many fragments landed on each gene. *Yield:* a **count table** — a grid of "gene × sample" with numbers. Higher number = that gene was more active in that sample.

Differential expression (DESeq2). Compare counts in diabetic vs healthy and ask, statistically, "*which genes are reliably different?*" A gene must be different *consistently*, beyond random noise. *Yield:* a table of **differentially expressed genes (DEGs)**, each with a fold-change (how much it moved) and a p-value (how confident). The **volcano plot** is just a picture of this table.

Splitting mRNA vs lncRNA (biotype step). Sort the changed genes: which are protein-coding (mRNA) and which are lncRNA. *Yield:* two clean lists — **DE mRNAs** and **DE lncRNAs** — the outer layers of the network.

Enrichment (clusterProfiler). Ask "*what do these changed genes have in common?*" (e.g. "many are involved in insulin secretion"). *Yield:* the biological *story* — which pathways are disturbed.

miRNA prediction (multiMiR / starBase). Standard RNA-seq can't see tiny miRNAs directly, so use databases that already know "miRNA A targets mRNA B" and "lncRNA C soaks up miRNA A." *Yield:* two connection lists (the network's wires).

Building + drawing the network (igraph / Cytoscape). Join the two wire-lists wherever they share the same miRNA, forming full lncRNA → miRNA → mRNA chains. *Yield:* the **ceRNA network**. The most-connected dots ("hubs") are your candidate master-regulators of diabetes.

One honest note for interviews

miRNAs here come from *prediction databases*, not your own sequencing. That's standard, but be ready to say:

"Standard RNA-seq doesn't capture mature miRNAs; they need small-RNA-seq. So I built the miRNA layer from curated interaction databases, and the natural next step would be matched small-RNA-seq to confirm them experimentally."

FAQ (your follow-up questions)

Q1. In the counting step, "gene X has 500 reads in sample 1, 1200 in sample 2" — what is a "sample"? Is it cells?

A "sample" is **one biological specimen** — here, the islet material taken from **one donor (one person)**. It is *not* a single cell.

This dataset is **bulk RNA-seq**: the RNA from *all* the cells in that person's islet material is mixed together in one tube and measured as a group. So the number you get is the **average activity across thousands/millions of cells** from that one donor. "Sample 1" and "sample 2" are **two** different people (say, one **healthy** and one **diabetic**). "Gene X = 500 in sample 1, 1200 in sample 2" means gene X was about twice as active in donor 2 as in donor 1.

(There's another kind called **single-cell RNA-seq** that measures each individual cell separately — but that's not what you're doing here. In your project, **sample = donor**.)

Q2. In DESeq2, what does "differentially expressed" mean?

Break the phrase in two:

- **"Expressed"** = how **active** a gene is / **how much RNA it's making**. High counts = **highly** expressed. Low counts = **barely** expressed.
- **"Differentially"** = **the difference between your two groups** (diabetic vs healthy).

So a **differentially expressed gene** is **one whose activity level is reliably different in diabetics than in healthy people** — and "reliably" is the **key word**. DESeq2 **doesn't just check** if the **average looks higher**; it **checks whether the difference is consistent across the donors** and **bigger than random noise**. A gene can be differentially expressed in **either direction**: **up** (more active in diabetics) or **down** (less active in diabetics). **Both count**.

Q3. In enrichment, what do "300 changed genes" mean — is it the differentially expressed genes, the high ones or the low ones?

"300 changed genes" is exactly your **differentially expressed gene list from DESeq2** — the genes that came out as **reliably different**. ("Changed" just means "changed compared to healthy.")

Importantly, it includes **both directions**: genes that went **up** (higher reads in diabetics) *and* genes that went **down** (lower reads in diabetics). It's not only the high ones or only the low ones — it's **every gene** that moved **significantly, up or down**.

Enrichment then takes that **whole list** and looks for **common themes** ("a lot of these genes are involved in insulin secretion"). Tip: **it's often insightful to run enrichment separately on the up genes and the down genes**, because "**what's being switched on in diabetes**" and "**what's being switched off**" can tell different stories.